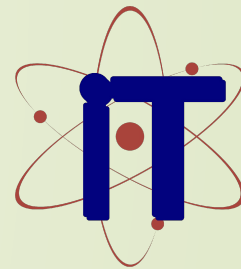




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Presentation on Diagrams NetAgent AI-Powered Network Security System

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1. Use Case Diagram

2

This use case diagram shows how different users interact with the **Net Agent AI-Powered Network Security System** to monitor, analyze, and secure a network.

Actors

1. Security Analyst / User

- Starts scans and monitoring.
- Views results and reports.
- Asks questions about network activity.
- Takes security actions based on recommendations.

2. System Administrator

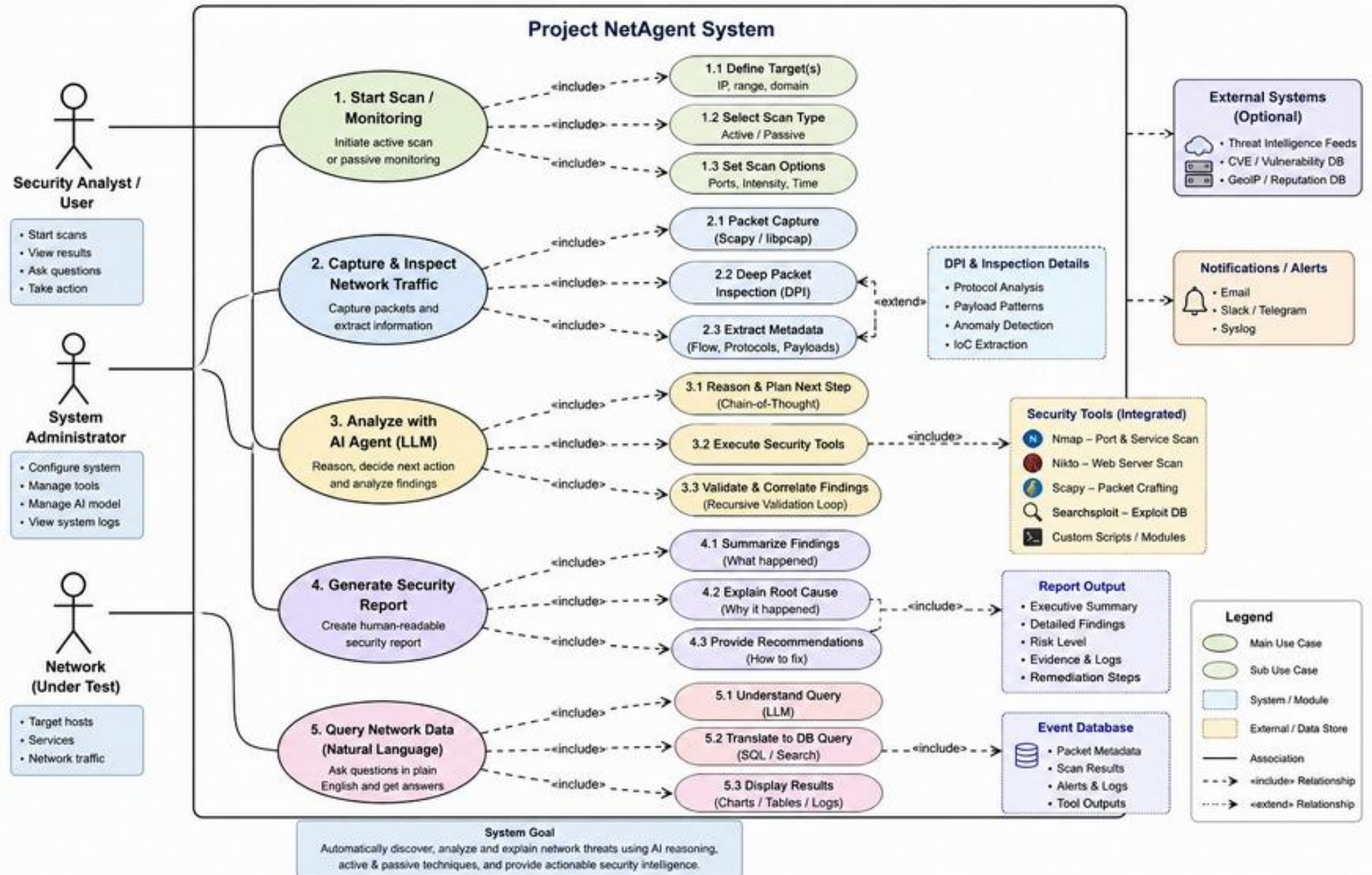
- Configures the system.
- Manages tools and AI models.
- Monitors system logs and performance.

3. Network (Under Test)

- The target network being scanned and monitored.

Use Case Diagram – Project NetAgent

AI-Powered Network Security Sentinel



2. Class Diagram

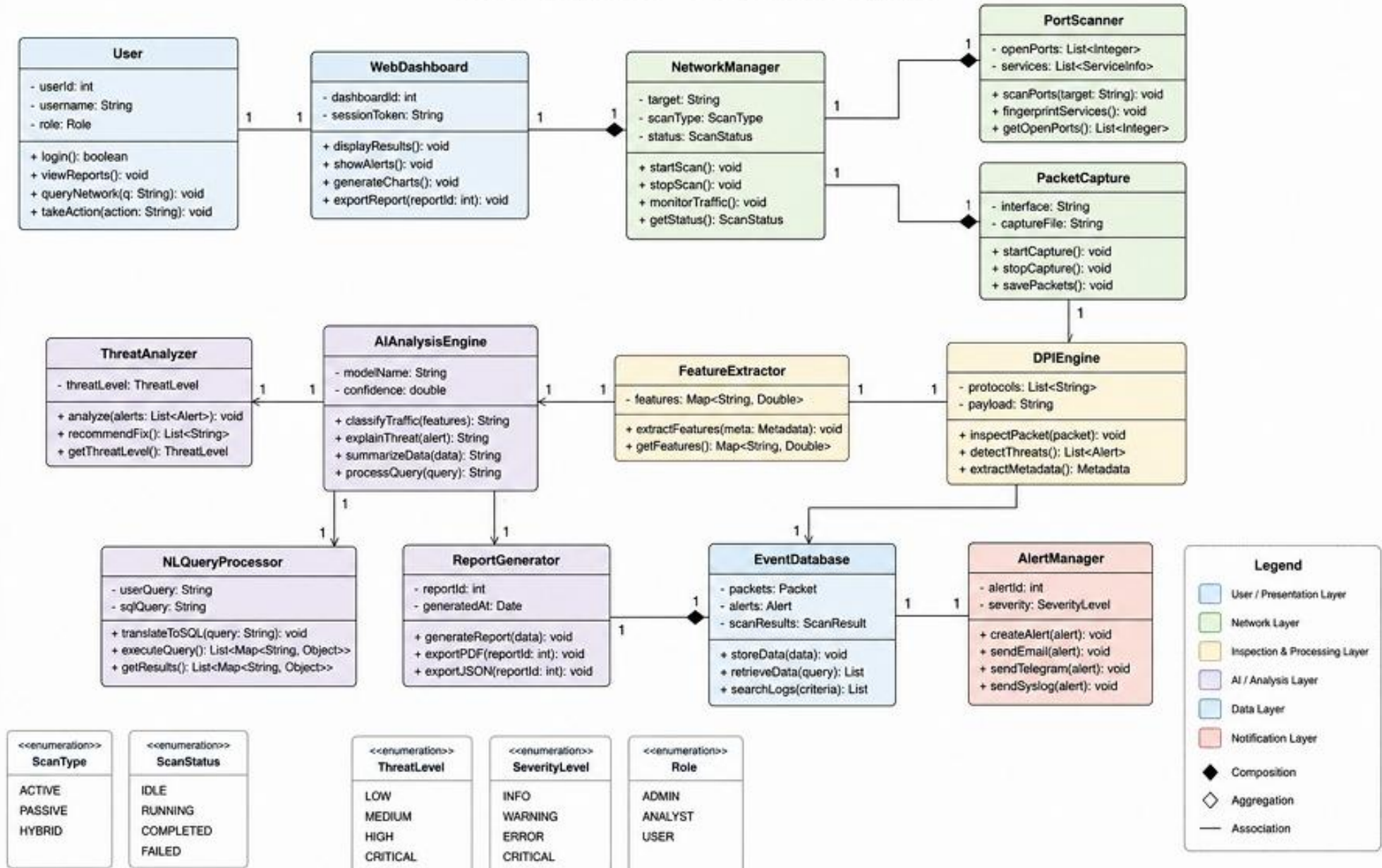
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The class diagram shows how NetAgent captures network traffic, analyzes it using AI, stores the results in a database, and provides reports and alerts to help users detect and respond to cybersecurity threats.

Relationship Summary

- User interacts with Web Dashboard
- Web Dashboard controls Network Manager
- Network Manager uses Port Scanner and Packet Capture
- Packet Capture sends traffic to DPI Engine
- DPI Engine uses Feature Extractor
- Feature Extractor sends data to AI Analysis Engine
- AI Analysis Engine works with:
 - Threat Analyzer
 - NL Query Processor
 - Report Generator
- All modules store/retrieve data from Event Database
- **Alert Manager** generates notifications from detected threats

Class Diagram – Project NetAgent



3. Activity Diagram

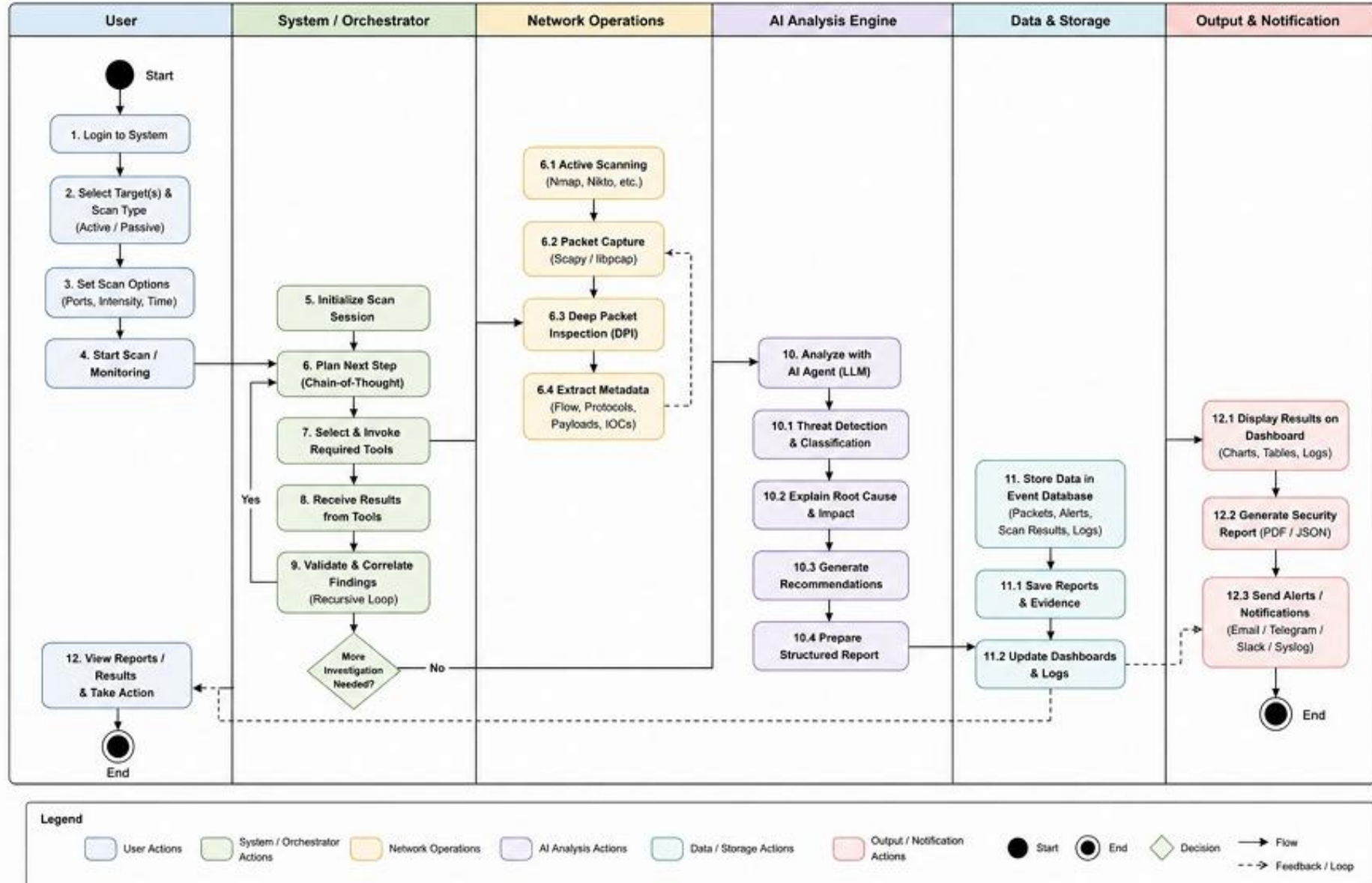
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The activity diagram shows how NetAgent automatically collects network data, uses AI for analysis, stores findings, generates reports/alerts, and supports user threat response.

1. User Starts Process
The user: Logs in -> Selects targets/networks -> Chooses scan type (Active/Passive) -> Sets options -> Starts scan/monitor.
2. System Initializes Scan
The system: Creates a scan session -> Plans actions -> Selects tools -> Collects results -> Validates/correlates findings. Loops for additional investigation.
3. Network Operations
The system performs monitoring: Active scanning (Nmap/Nikto) -> Packet capture (Scapy/libpcap) -> Deep Packet Inspection (DPI) -> Metadata extraction (protocols, payloads, IPs, IOCs). This provides raw data.
4. AI Analysis
The AI Agent (LLM) analyzes data: Detects threats -> Classifies severity -> Explains root cause -> Assesses impact -> Generates security recommendations -> Prepares a structured report.
5. Data Storage
All information is stored in the Event Database: Packets -> Alerts -> Scan results -> Logs -> Evidence. Dashboards and records are updated.
6. Output and Notifications
The system: Displays results on the dashboard -> Generates PDF/JSON reports -> Sends alerts via Email, Telegram, Slack, or Syslog.
7. User Reviews Results
The user: Views reports/findings -> Reviews recommendations -> Takes corrective actions. The process ends.

Activity Diagram – Project NetAgent

AI-Powered Network Security Sentinel



4. Sequence Diagram

8

The sequence diagram illustrates how a scan request travels through NetAgent—from user input, data collection, and AI-based threat analysis, to storing results, generating reports, sending alerts, and finally presenting the findings to the user.

i. Scan Setup

The process starts when the User:

- Logs into the system
- Configures scan settings (targets, scan type, options)
- Starts the scan The Web Dashboard sends this request to the (System Manager).

ii. Data Collection

The **Orchestrator**:

- Creates a scan session
- Initializes security tools and resources
- Starts network operations

The Network Operations module then:

- Performs port scanning
- Detects services
- Captures packets
- Performs Deep Packet Inspection (DPI)
- Extracts metadata The collected results are sent back to the Orchestrator.

iii. AI Analysis

9

The Orchestrator sends the collected data to the AI Analysis Engine (LLM). The AI then:

- Detects threats
- Classifies attack severity
- Analyzes root causes
- Assesses impact
- Generates recommendations
- Prepares a structured security report

The analysis results are returned to the Orchestrator.

iv. Storage and Reporting

The Orchestrator:

- Stores packets, logs, alerts, and findings in the **Event Database**
- Requests the **Report Generator** to create reports
- Sends alerts to the **Alert Manager**

The Alert Manager delivers notifications through:

- Email
- Telegram

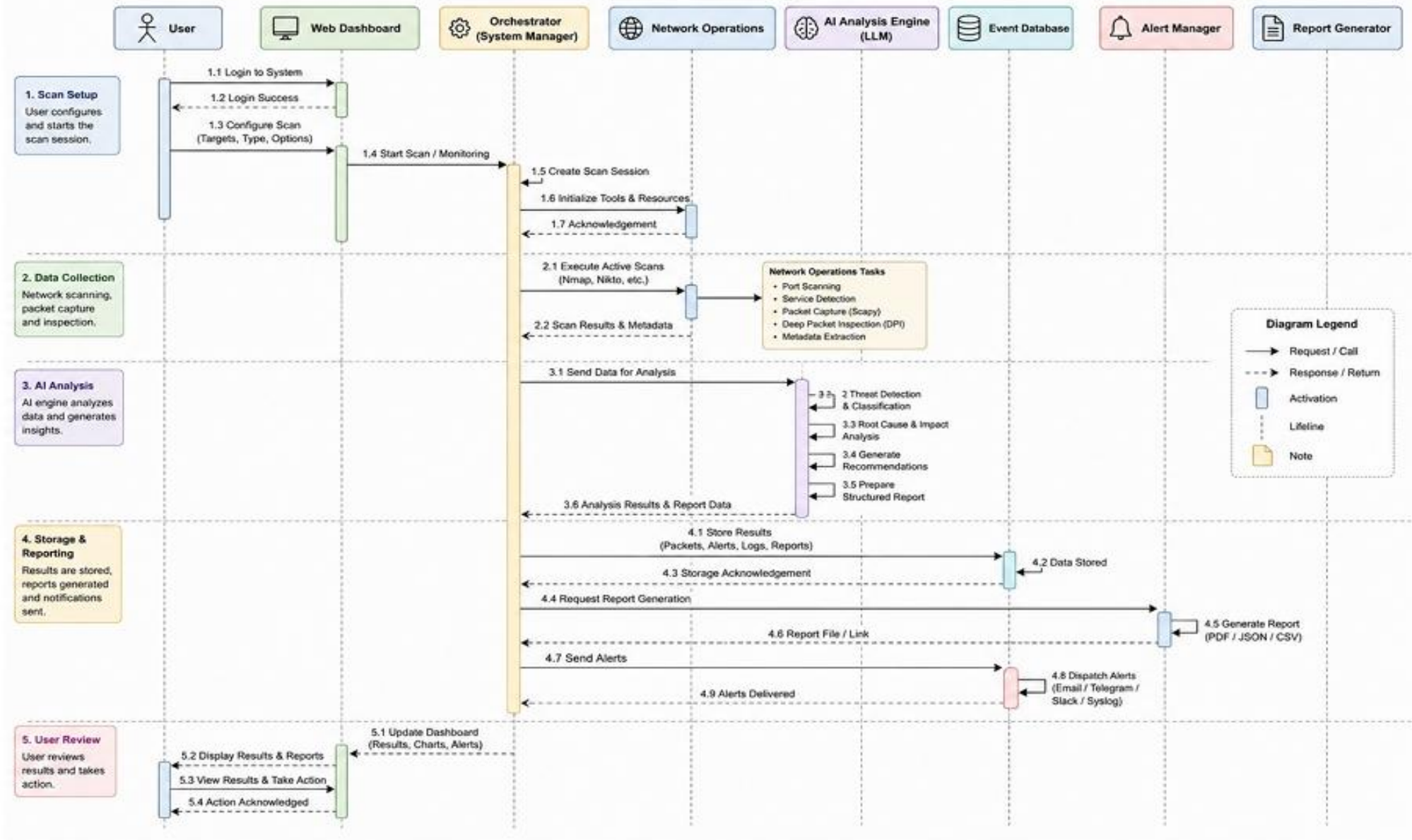
v. User Review

Finally:

- The Web Dashboard updates with the latest results
- Reports, charts, and alerts are displayed
- The User reviews the findings and takes action if required The system then acknowledges action and the process ends.

Sequence Diagram – Project NetAgent

AI-Powered Network Security Sentinel



5. State Diagram

11

The state diagram shows how NetAgent moves from waiting for a scan, to collecting and analyzing network data, generating reports and alerts, storing results, and finally ending or archiving the session.

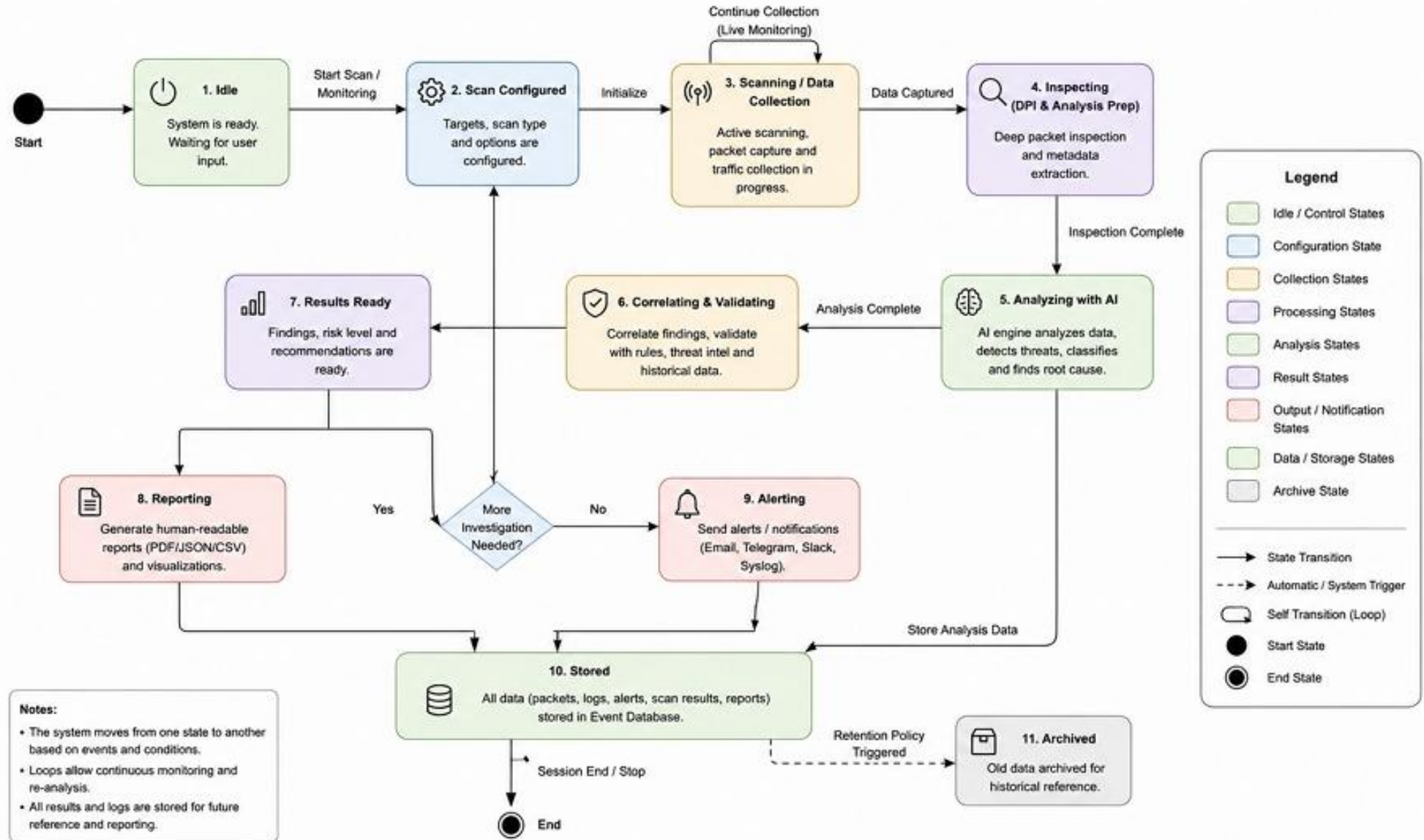
A state diagram shows the different states that NetAgent goes through while performing network monitoring and threat analysis.

Flow of States

- **Idle:** The system is waiting for the user to start a scan.
- **Scan Configured:** The user selects targets and scan settings.
- **Scanning / Data Collection:** NetAgent scans the network and captures traffic data.
- **Inspecting:** The system performs Deep Packet Inspection (DPI) and extracts information.
- **Analyzing with AI:** The AI examines the collected data to detect threats and identify activities.
- **Correlating & Validating:** Findings are checked against threat intelligence, rules, and previous data to confirm accuracy.
- **Results Ready:** The system prepares the final findings and recommendations.
- **Reporting:** Security reports are generated for the user.
- **Alerting:** Notifications are sent if threats or risks are detected.
- **Stored:** All logs, packets, alerts, and reports are saved in the database.
- **Archived:** Old data is moved to storage for future reference.
- **End:** The monitoring session finishes.

State Diagram – Project NetAgent

AI-Powered Network Security Sentinel



6. Data Flow Diagram

13

The Data Flow Diagram shows how NetAgent collects network data, analyzes it using AI, stores the results, and delivers reports and alerts back to the user.

1. User Provides Input

- The user enters scan settings, target IPs, or asks network-related questions.
- This information is sent to the NetAgent system.

2. Network Data Collection

- NetAgent scans the network and captures packets.
- Data is collected from target devices and network traffic.

3. Traffic Inspection

- The system performs Deep Packet Inspection (DPI).
- Important information such as protocols, ports, IPs, and payload details is extracted.

4. AI Analysis

- The extracted data is sent to the AI Engine.
- The AI identifies threats, analyzes risks, and generates recommendations.

5. Data Storage

- Scan results, logs, packets, and alerts are stored in the Event Database.
- The database acts as the central repository for all collected information.

6. Report & Alert Generation

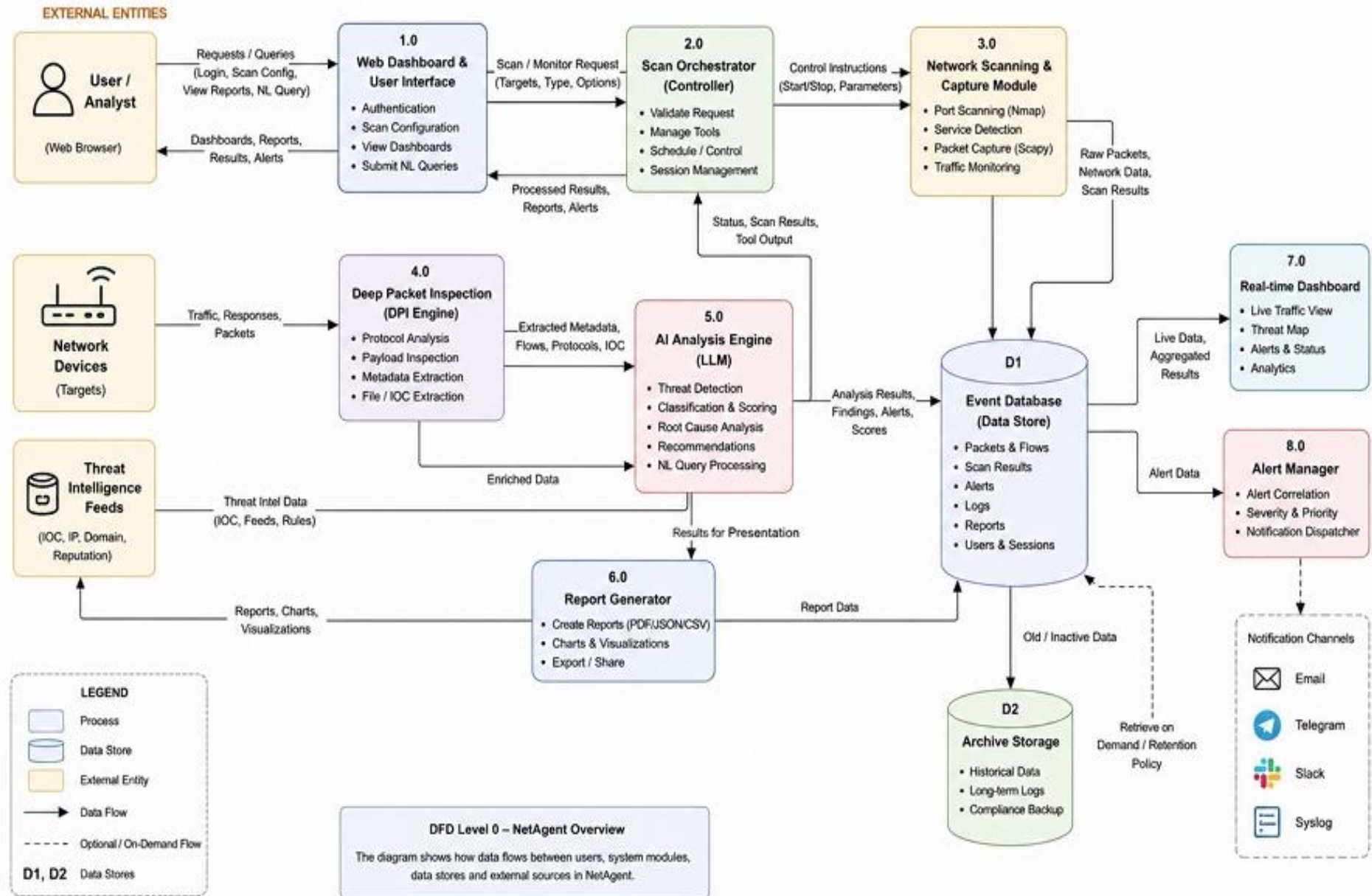
- The Report Generator creates security reports.
- The Alert Manager sends notifications if suspicious activity is detected.

7. Results to User

- Reports, alerts, charts, and recommendations are displayed on the dashboard.
- The user reviews the findings and takes action if needed.

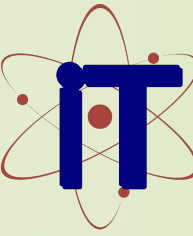
Data Flow Diagram (DFD) – Project NetAgent

AI-Powered Network Security Sentinel





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Thank You
Queries ?